

SQL Data Quality & Star Schema Modeling Project

1. Project Overview

Database: Oracle SQL

Objective: Clean raw operational tables, enforce data integrity, and model data using star schema.

Tools: Oracle SQL Developer, dbdiagram.io

2. Data Connection

Connection used: Oracle SQL Developer 23.x

Local DB with SID: ORCL, user: ecom_user

3. Data Cleaning & Constraints

- Removed duplicates in 'products' table using ROWID
- Created surrogate primary key for 'order_items'
- Enforced primary/foreign keys across all operational tables

4. Star Schema Design

Fact Table: fact_sales (order_id, customer_id, product_id, order_date_key, delivery_date_key, payment_value)

Dimension Tables: dim_customer, dim_product, dim_date, dim_order_status

Generated 'dim_date' using PL/SQL loop with date range

5. ERD & Modeling

ERD designed using dbdiagram.io and exported for presentation

Star schema modeled with clear one-to-many relationships

6. Analytical Query Examples

- Top cities by sales
- Monthly sales trend
- Product-level revenue analysis

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7. Summary for Resume

- Cleaned and normalized raw e-commerce data in Oracle
- Built a robust star schema with fact and dimension tables
- Enforced data quality via PK/FK constraints
- Created custom calendar table (dim_date) for time analysis
- Designed and shared ERD using dbdiagram.io
- Wrote complex SQL for BI analysis (monthly sales, top customers)